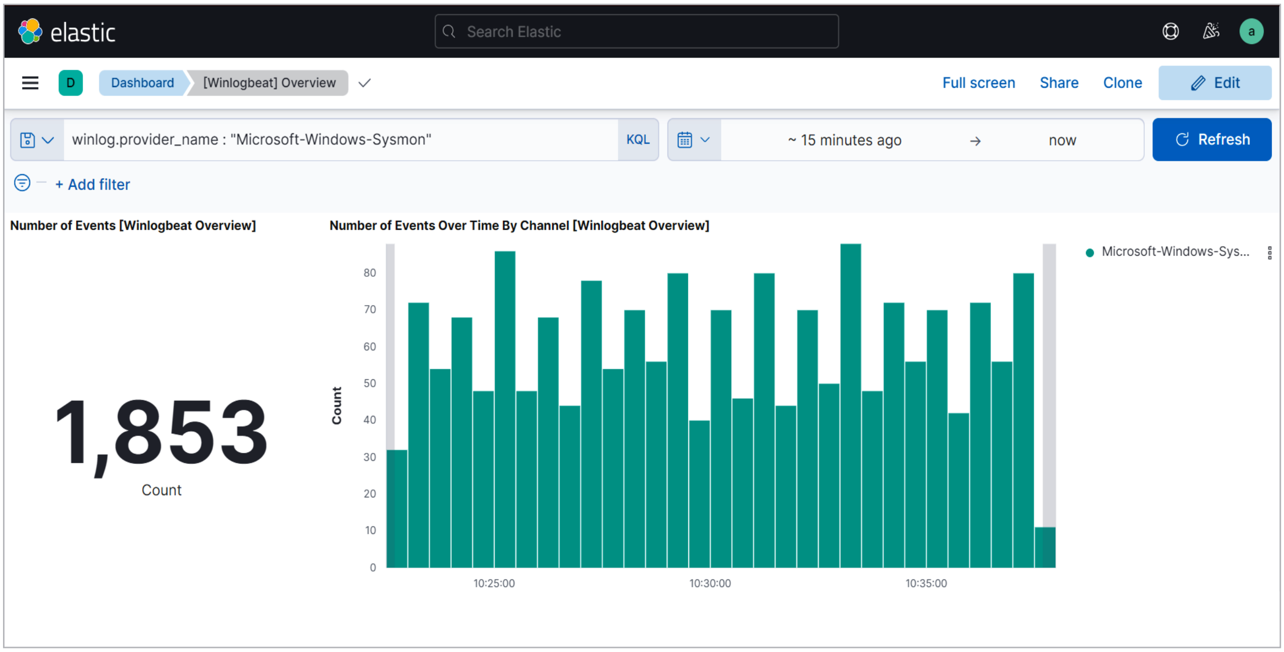

Figure 7: Sysmon logs in Kibana

Sysmon

System Monitor (Sysmon) is a Windows service. Once installed and configured it remains persistent, and logs system activity to the Windows event log. This module provides very detailed information about process creation, network connections and any file changes.

Installation of Sysmon

For complete setup of Sysmon, two things must be installed. One is a Sysmon tool available in the zip file and the other is the configuration file from GitHub. The download links are given below:

- *https://docs.microsoft.com/en-us/sysinternals/downloads/Sysmon*
- *https://github.com/SwiftOnSecurity/Sysmon-config*

Create a new folder in *C:/Program Files/*. Extract both the files in that folder. Now open PowerShell as administrator and navigate to that directory. Run the following command:

```
.\sysmon64.exe -accepteula -i
sysmonconfig-export.xml
```

Open *Local Group Policy* (search in *Start*), and navigate to *Computer Configuration > Policies > Administrative Templates > Windows Components > Windows PowerShell*.

On the right side, enable 'Turn on Module Logging' and 'Turn on PowerShell Script Block Logging' (refer to Figure 6).

Now we need to update Winlogbeat so that it ships the logs from Sysmon too. For that, once again open the *winlogbeat.yml* file. Under the *winlogbeat.event_logs* section, make sure the following are added:

- *Microsoft-Windows-PowerShell/Operational*
- *Microsoft-Windows-Sysmon/Operational*

Once this is done, let's check whether the logs are being sent to Kibana by Winlogbeat or not. Enter the following search query to check:

.

```
winlog.provider_name : "Microsoft-
Windows-Sysmon"
```

The result of this query gives the details of logs sent by Sysmon (refer to Figure 7). If you didn't find any logs, try to increase the time period or generate some traffic.

This is how we can install and configure Winlogbeat and Sysmon to stream logs to Elasticsearch and visualise the data. Depending on your use cases, you can add the name of the event logs to be sent in *winlogbeat.yml* file and cutomise the logs visualisation as per your requirement. END

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Both the authors like to work in the field of network analysis, IoT and security.